Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Roots of Polynomials

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. The cubic $x^3 + ax^2 + bx + c = 0$ has roots α, β, γ. Write down $\alpha + \beta + \gamma$. **[1]**

2. For the same cubic, write down $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **[2]**

3. The equation $x^3 - 6x^2 + 11x - 6 = 0$ has roots $1, 2, 3$. Verify the sum and product of roots. **[3]**

4. Find a cubic equation with roots $1, 2, 4$. **[3]**

5. The quadratic $x^2 - 5x + 6 = 0$ has roots α, β. Find $\alpha^2 + \beta^2$. **[3]**

6. For roots α, β of $x^2 - px + q = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$. **[3]**

7. A cubic has roots $2, -1, 3$. Find the equation. **[3]**

8. If α, β, γ are roots of $x^3 - 3x^2 + 2x - 1 = 0$, find $\alpha + \beta + \gamma$. **[2]**

9. Find $\sum \alpha^2$ for the cubic $x^3 - 3x^2 + 2x - 1 = 0$. **[4]**

10. The equation $x^3 + px^2 + qx + r = 0$ has roots α, β, γ. Express p, q, r in terms of the roots. **[3]**

11. One root of $x^3 - 4x^2 + x + 6 = 0$ is $x = -1$. Find the other roots. **[4]**

12. Form the equation with roots $2\alpha, 2\beta, 2\gamma$ where α, β, γ are roots of $x^3 - x^2 + 2x - 3 = 0$. **[4]**

13. If $\alpha + \beta = 5$ and $\alpha\beta = 6$, find the quadratic equation. **[2]**

14. Find $\alpha^3 + \beta^3$ given $\alpha + \beta = 3$, $\alpha\beta = 2$. **[3]**

15. State the factor theorem. **[2]**

Section B – Hard (Q16–25)

16. Roots α, β, γ of $x^3 - 6x^2 + 11x - 6 = 0$. Find the equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. **[5]**

17. Find the equation with roots $\alpha + 1, \beta + 1, \gamma + 1$ for the same cubic. **[5]**

18. Given α, β, γ roots of $x^3 + ax^2 + bx + c = 0$, find $\sum \alpha^2\beta$. **[4]**

19. Solve $x^3 - 3x + 2 = 0$ given that one root is 1. **[4]**

20. The cubic $x^3 - 7x - 6 = 0$ has roots α, β, γ. Find $\alpha^3 + \beta^3 + \gamma^3$. **[4]**

21. Form the quadratic with roots α^2, β^2 where α, β are roots of $x^2 - 3x + 1 = 0$. **[5]**

22. A polynomial $p(x)$ has $(x - 2)$ as a factor and $p(1) = 0$. Write two linear factors. **[2]**

23. Find the condition on a, b, c so that $x^3 + ax^2 + bx + c = 0$ has a repeated root. **[5]**

24. Roots α, β, γ satisfy $\alpha + \beta + \gamma = 0$. Show that $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$. **[4]**

25. Find a cubic equation with integer coefficients having roots $1 + \sqrt{2}$ and $1 - \sqrt{2}$, and one other integer root of your choice. **[5]**

End of Worksheet • *Show all working clearly.*